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Optical logic operations using a meshed Mach-Zehnder interference array

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ABSTRACT

Optical logic operations play a significant role in digital optical computing and signal processing. Conventional logic operations always rely on nonlinear light-matter interactions, where high energy consumption, restricted logical function, and poor cascability seriously hamper their practical application. In this study, we introduce a meshed Mach-Zehnder interferometer (MZI) array to provide a complex linear-modulation solution for optical logic operations. The optical interference unit (OIU) is formed by a meshed MZI array, and the unitary matrix-vector product operations are imitated. The introduction of a linear vector composition makes it possible to optimize the phase-shift parameters of the OIU using a gradient descent algorithm for non-unitary logic operations. In the simulation, we demonstrate that seven basic logic operations are well implemented by an OIU with three MZIs. Complex logic operations and multi-input logic operations are also achieved by cascading basic components and increasing the size of the OIU. These linear-optical logic operations are implemented on the same platform using phase-only modulation, which has low power consumption and is critical for device fabrication and system integration.

1. Introduction

The development of optical digital computing and signal processing requires high-speed and low-power optical logic operations [1–4]. Since logical variables frequently exhibit nonlinear transform relationships, optical nonlinearities have become a common solution for implementing logical functions. The optical nonlinearities of materials involving cross-gain modulation and four-wave mixing have been used to produce logic gates [5–12], and diverse micro-nano structures have been proposed to enhance the nonlinear effect, such as photonic crystal nanocavities and micro-ring resonators [13–21]. However, nonlinearities are typically dependent on optical power and micro-ring structure, and different logic operations require different nonlinear properties, making it difficult to perform all basic logic operations in a single framework and to realize complex logic operations with multiple inputs. In addition, the intensity-dependent inhomogeneous transmittance during nonlinear optical parameter conversion will not only result in a mismatch between the input and output states, but also leads to high power consumption, which are the obstacle to basic operations and logic functions. The two most important problems in the practical application of optical digital

computing and signal processing are complete basic logic gates and their cascability; however, they remain unresolved. Therefore, it is urgent to explore new approach to realize optical logic operations and overcome the aforementioned obstacles.

In this study, we introduce a meshed Mach-Zehnder interferometer (MZI) array to provide a complex linear modulation solution for optical logic operations. The optical interference unit (OIU) was constructed using a meshed MZI array, and unitary matrix transformations were mimicked by multi-beam interference [22,23]. Although MZI structures have previously been used to perform logic operations, nonlinear factors must be introduced because the MZI can only control the light field unitarily [24–28]. In addition, optical nonlinear effects dependent on optical power require precise amplitude modulation to ensure logical contrast, which degrades the robustness of logic gates and restricts their cascability. Although some plasmonic and MZI-based linear switching devices have been proposed to achieve basic logic operations [29–35], they are difficult to implement complex multi-input logic operations, and the design of the architecture often relies on fixed empirical paradigms. Inspired by linear algebraic composition, we observed that logic operations can be achieved by local interference with a complex linear

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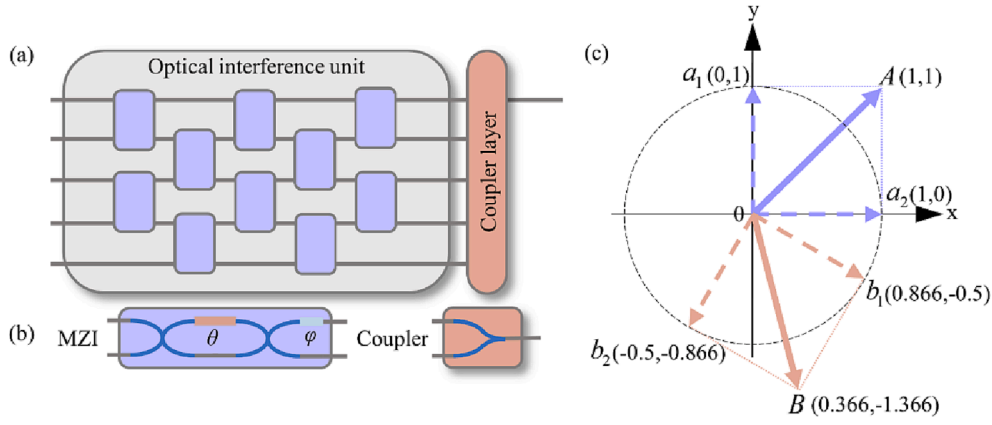


Fig. 1. Schematic of MZI array used to perform logic operations. (a) OIU composed of MZI array for unitary transformation and coupler layer composed of multiple-to-one coupler for linear vector composition. (b) Structures of a programmable MZI and a coupler. (c) Unitary orthogonal composition diagram of two-dimensional vectors.

modulation, which is characterized by a unitary OIU. Furthermore, its phase-shift parameters can be optimized using gradient descent algorithms [36,37], which provide a promising solution for the linear composition of optical logic operations.

Because the logic operations are performed by linearly modulating light fields, they do not require high optical power and can maintain the power relationship between the output and input logical states. Because the logic operations manipulate the normalized amplitude end-to-end, they are power-independent, and robust performance can be guaranteed. Consequently, we constructed a 3-port OIU and tested its capability to perform logical operations. The simulation results show that the OIU with three MZIs can perform all seven basic logic operations, and complex logic operations, such as half-adder or multi-logic operations, are also performed by cascading the basic components. Furthermore, multi-input logic operations, including a 4-bit odd-even discriminator, were implemented by extending the OIU size. In addition, the output power of each logic operation changes approximately linearly with the input power, and the output logical contrast remains unchanged, indicating strong stability and a great fanout. Because these logical operations are performed on the same platform, this strategy may provide a general and promising method for the integration of devices and systems, which may have broad applications in optical digital computing and signal processing.

2. Principles and methods

2.1. Logic operations using Mach-Zehnder interference array

Fig. 1(a) shows the schematic of the optical logic operations implemented using the OIU, including the MZI array and the coupler layer. The logic state vector was modulated at the amplitude of the optical pulses and input into the OIU. The MZI array applied a unitary transformation to the input logic vector, and the coupler layer superimposed the transformation outputs linearly to produce a single-output logic state. The MZI contained two 50:50 beam splitters (dark blue cross lines in Fig. 1(b)) and two phase shifters parameterized by ϕ and θ (brown and light blue bars in Fig. 1(b)). Here, we assumed that the MZI is lossless and can implement a 2×2 unitary transform on the input states with modes (k_1, k_2) into the output states with modes (k'_1, k'_2) , which can be represented as follows [22,38]:

$$\begin{bmatrix} k'_1 \\ k'_2 \end{bmatrix} = \begin{bmatrix} e^{i\phi} \sin \theta & e^{i\phi} \cos \theta \\ \cos \theta & -\sin \theta \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}. \quad (1)$$

Therefore, the coupling of the two optical signals in the MZI can be changed by modulating (ϕ, θ) to achieve the desired linear complex

amplitude modulation, which is called a unitary transformation. By successively applying different forms of unitary operators, an OIU with N input ports can perform an $N \times N$ unitary transformation of the input states with modes of $(k_1, \dots, k_n)$ into the output states with modes of $(k'_1, \dots, k'_n)$. The unitary operators in the OIU are constructed according to the principles defined by Clements *et al.* [39], where the MZI that between the ports of i and j performs the transformation as follows:

$$u_{ij}(\phi, \theta) = \begin{bmatrix} 1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & e^{i\phi} \cos \theta & -\sin \theta & \dots & 0 \\ 0 & \dots & e^{i\phi} \sin \theta & \cos \theta & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & \dots & 1 \end{bmatrix}, \quad (2)$$

Therefore, any unitary matrix performed by OIU can be written as follows:

$$U = D \left(\prod_{ij \in S} u_{ij} \right), \quad (3)$$

where S represents the specific ordered sequence for MZIs within the OIU and D represents a diagonal matrix [39]. Note that the form of unitary transformation here specifies the transformation that the OIU can perform, where the output state corresponding to the input state is determined by ϕ and θ . The output state for an arbitrary input state K is expressed as $K' = UK$, where K and K' represent N -dimensional column vectors. Furthermore, we introduce the coupler layer to combine the results of unitary matrix-vector multiplication to obtain the final output that represents the logic output, which can be mathematically expressed as the sum of some elements of K' . The introduction of a coupler layer can destroy the unitary containment in the MZI array, satisfying mapping relationships when performing optical logic operations.

Light fields are generally characterized by complex amplitudes, and logical variables are represented by amplitude or optical intensity (the square of the amplitude), where thresholds are set to divide logical states "0" and "1" for binary logic operations. In linear optics, the output light field can be expressed as a linear superposition of all input fields and can be written as a vector product $Y = AX$, where Y represents the logic output, A represents the modulation coefficient, which is a row vector, X is the input logic containing binary variables, which is a column vector. Different logic operations can be represented by changing the coefficient vector of A , where the modulus square of the coefficient matrix is an essential factor that determines the form of the coupler layer. A detailed mathematical analysis is provided in the [Supplementary Information I](#). For seven basic logic operations with two logic inputs and one logic output, the square modulus of the vector A is constrained by $|A|^2 \in [1, 2]$,

3]. For instance, when the modulo square of $|A|^2 = 1$ for the AND operation, it is 2 for the OR operation. Because the unitary matrix defined in Eq.(3) requires all column (row) vectors to be standard orthogonal unit vectors, the unitary matrix has difficulty achieving all logic operations, and the corresponding transformations required are usually non-unitary and cannot be represented by unit vectors [34]. Therefore, implementing seven basic optical logic operations necessitates simultaneous linear unitary transformation and coupled superposition of the inputs. We map the logic transformations into multiple orthogonal vector combinations of U by introducing a multiple-to-one direction coupler at the backend of the OIU. A is represented by a linear superposition of multiple unit vectors of U . The operator C represents the linear vector composition introduced by the coupler, and its transmission matrix is based on the specific logic operation, which can be expressed as an N -dimensional row vector containing only 0 and 1. Fig. 1c shows an example of a two-dimensional case: the unitary orthogonal composition of the two-dimensional vector introduced by the coupler with $C = [1, 1]$. Any vector with a modulus square of two can be linearly composed of two orthogonal unit row vectors of a two-dimensional unitary matrix (e.g., $A = a_1 + a_2, B = b_1 + b_2$).

For a matrix composed of all possible input states, the logic operation realized by the combined action of the MZI array and coupler layer can be expressed as follows:

$$Y = CUX \quad (4)$$

where X now represents an $M \times N$ -dimensional matrix with each column representing a possible input state, and Y represents an M -dimensional vector with each element representing a corresponding output state. For any N -dimensional unitary matrix, the maximum modulus square of the vector that can be composed by linearly superimposing the row vectors is equal to N . Therefore, when X and Y represent the input set and corresponding output set of the logic operation, respectively, the desired logic operation can be implemented by solving the modulated parameters carried by the operator $A = CU$.

2.2. Gradient descent algorithm-assisted parameters optimization

To solve the desired unitary transformation U and coupler operation C matrices that satisfy the transformation mapping from X to Y , we introduce a gradient descent algorithm to optimize the matrices. Although the desired inputs and outputs of X and Y are determined and an analytical solution that satisfies Eq.(4) exists, it is difficult to obtain an algebraic solution constructed by the MZI array that performs matrix transformation, particularly for complex logic operations. Additionally, because the mapping relationship between each input and output state is not always linearly independent, the corresponding linear equations may have no solution, indicating that the rank of the coefficient matrix A may not be equal to the rank of the augmented matrix $[A, Y]$. However, we can turn these constraints and trade-offs into an optimal solution problem using an algorithm to determine the optimal algebraic solution of the OIU. We introduce a gradient descent algorithm to the target-oriented solution of the phase-shift parameters of the MZI array and fit the mapping related to the logic operation of the target [36]. The optimization target iteratively minimizes the objective function, which can be described as follows:

$$\min_{\phi, \theta} L(Y, \tilde{Y}), \quad (5)$$

where $L(\cdot)$ is designed to calculate the target output Y and to predict the output $\tilde{Y}$. When the optimization process converges, optimal (ϕ, θ) can be achieved to convert X to Y . Here, the objective function to be optimized is defined as follows:

$$L(Y, \tilde{Y}) = \frac{1}{n} \sum_i (Y^{(i)} - \tilde{Y}^{(i)})^2 \quad (6)$$

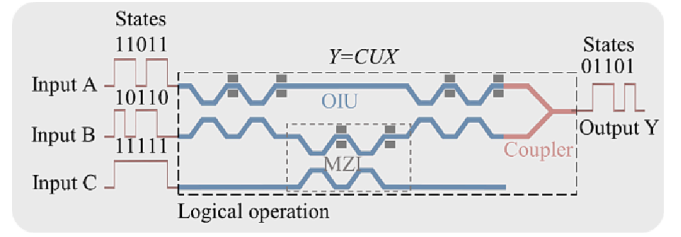


Fig. 2. Schematic of the optical logic operation using Mach-Zehnder interference array.

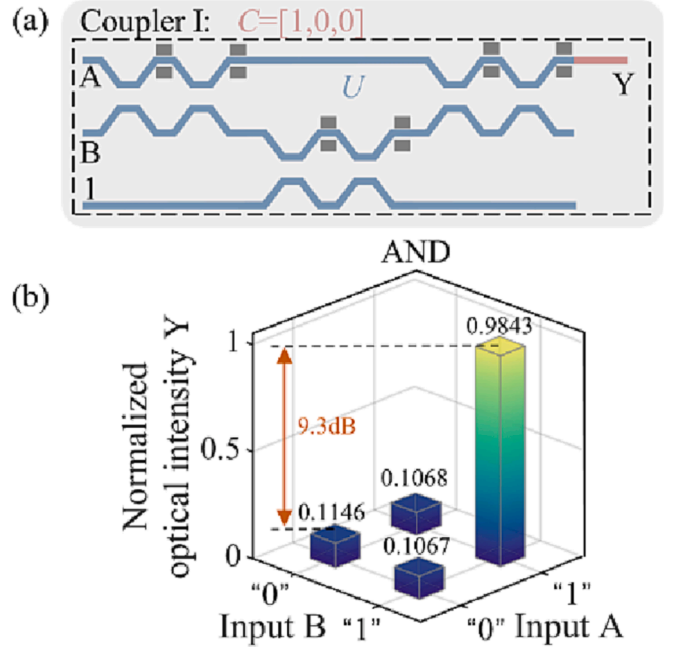


Fig. 3. Simulation results of AND operation with a three-port OIU. (a) The configuration of OIU and coupler. (b) The output normalized optical intensity of AND operations for input logic states of $(A, B) = (0,0), (1,0), (0,1),$ and $(1,1)$.

where n denotes the total number of possible samples. The process of optimizing the algorithm is described in [Supplementary Information II](#).

3. Results

Because the transformation matrix of the MZI is independent of incident optical intensity (amplitude), we utilize the normalized amplitude to characterize the logical variable, and the logical states of "0" and "1" are associated with the normalized amplitudes of 0 and 1, respectively. It is worth noting that the normalized amplitude of one was the maximum optical power when performing the experiment. Here, an OIU comprising three meshed MZIs and a coupler at the backend was used to implement the logic operations. The phase-shift distributions were solved by applying the gradient descent algorithm. Fig. 2 shows a schematic of the OR logic operation using the meshed MZI arrays, which contain two logic input ports with binary variables A and B and one invariant input port with constant C . The optical modes carrying logic information are first coupled into the OIU and then fully interfere within the array to realize the orthogonal vector mapping of logic operations in the unitary transformation. The calculation results of multiple orthogonal vectors are converted to logical output Y using a coupler with $C = [1, 1, 0]$. It allows the MZI array to perform complex amplitude modulation of the input optical modes by adjusting the phase-shift parameter

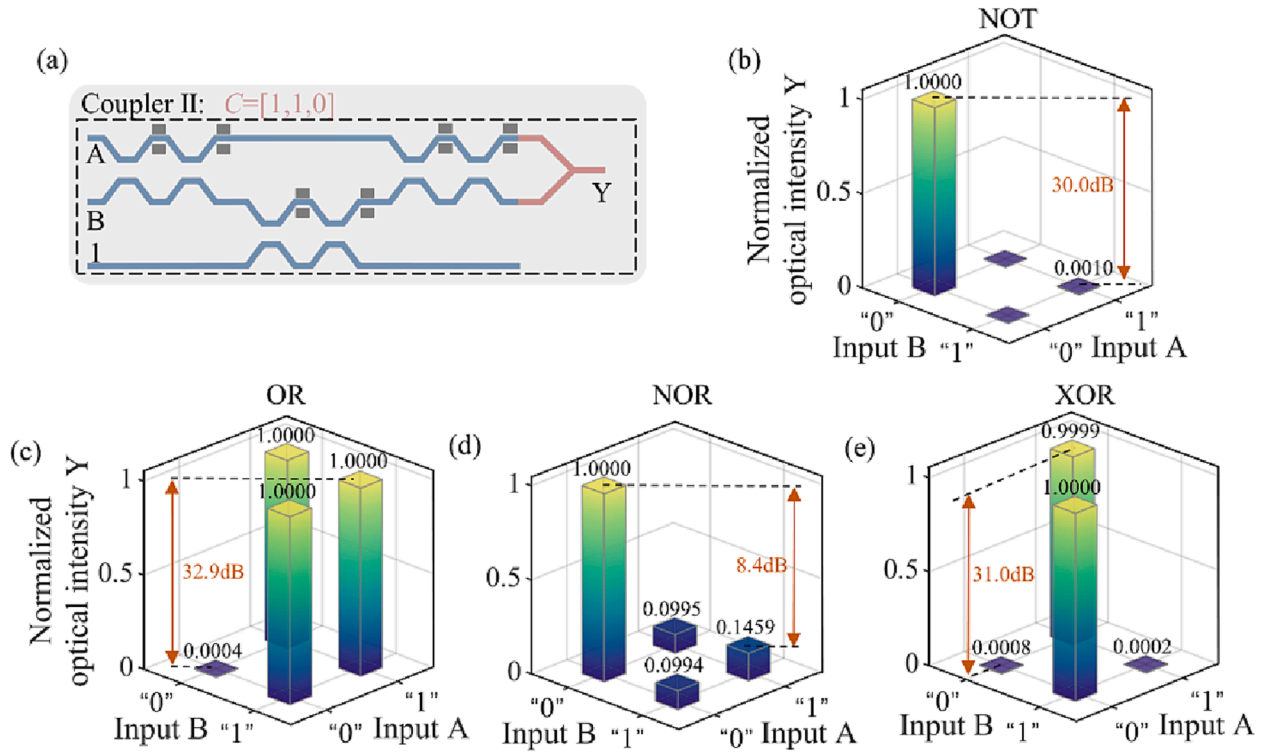


Fig. 4. Simulation results of NOT, OR, NOR, and XOR operations with a three-port OIU. (a) The configuration of OIU and coupler. (b) The output normalized optical intensity of (b) NOT, (c) OR, (d) NOR, and (e) XOR operations.

and realizing the logical state transformation between the input and output. The optical logic operations were implemented using Python version 3.6.1 and Tensorflow framework version 2.4.0 using a desktop computer equipped with a Core i9-10900X CPU and an NVIDIA GeForce RTX 2080Ti GPU. The phase shifters of logic gates are optimized by tensor-based gradient descent algorithm. The gradient descent algorithm has a learning rate of 0.001, and it took approximately 5 min for optimization.

3.1. Implementation of seven basic optical logic operations

Here, we first demonstrate the implementation of AND logic operations. The configuration of the OIU and coupler is shown in Fig. 3a. An ideal AND operation should provide “1” output intensity only when the two inputs A and B are “1”. For the input logic states $(A, B) = (0, 0), (0, 1), (1, 0), \text{ and } (1, 1)$, the AND operation returns 0, 0, 0, and 1, respectively. Fig. S1(a) presents the corresponding truth table. Because the modulo square of the modulation coefficient A is equal to 1, it can be represented by a single row vector of a unitary matrix, where coupler I is set as $C = [1, 0, 0]$ for the AND operation. Therefore, we consider the first output port of the OIU as the output of the AND operation, and discard the other ports. The OIU model for the AND logic operation was established by feeding the logic sequence of A and B into Eq. (4) to calculate the desired phase-shift parameters, where the loss function curve during optimization is shown in Fig. S4(a). During the iterative process, the loss value sharply drops at first and then stabilizes and converges at approximately 0.009. We use the normalized optical intensity (normalized amplitude squared) to characterize the output Y because the optical intensity is a detectable physical quantity. The x- and y-axes represent the input states of A and B, respectively, and they contain two values: “0” and “1,” representing normalized amplitude 0 and 1. As shown in Fig. 3(b), the output optical intensities corresponding to the four input logic states were 0.1146, 0.1067, 0.1068, and 0.9843, respectively. The minimum intensity contrast ratio (defined as the ratio between the maximum intensity of “0” and the minimum intensity of

“1”) between the output logic states “1” and “0” is 9.3 dB. Because of the squared relationship between the amplitude and optical intensity, the contrast ratio of the amplitude is equal to half of the intensity by approximately 4.7 dB. Because the intensity of the output logic state “0” is not zero, the AND operation is not ideal, but relatively optimal. This is also an inherent flaw in logic operations based on linear optics, and such a phenomenon also exists in logical NOR operations. This is because logic states “1” can be mapped to different points (vector) on the complex plane, while logical “0” can only be represented by the origin. Therefore, for logic operations that contain three “0” outputs and only one “1” output, we can only make a trade-off between contrasts. This issue is briefly discussed in [Supplementary Information I](#).

We then consider the implementation of other logic operations, including NOT, OR, NOR, and XOR operations. Fig. 4(a) shows the architecture of the OIU and coupler, where the settings of the input ports are the same as those of the AND. In contrast, coupler II is set as $C = [1, 1, 0]$ to introduce the local interference of the two outputs of the OIU, which mathematically enables the OIU to synthesize a vector with a modulo square of two by summing two standard orthogonal row vectors mathematically. Similarly, we substituted the input–output relationship that satisfied the logical operations of OR, XOR, and XNOR into Eq. (4) and combined with the gradient descent algorithm to reduce the loss function and optimize the phase-shift parameters in the OIU. Fig. 4(c)–(e) show the corresponding output intensities of the logical operations of OR, NOR, and XNOR. As shown in Fig. 4(c) and (e), the output intensities of OR and XOR operations show nearly ideal responses, the error between the normalized output intensities representing the logical states “1” and “0” and the ideal values are less than 0.001 with the logical contrast ratio greater than 30 dB. This indicates that this configuration fits well with the transformation matrix required for the logic operations. Fig. 4(b) shows the output of the NOT logic operation, which considers one input A and transforms it into the opposite state. For the input logic states of “0” and “1,” the corresponding output states are “1” and “0,” respectively. The results show that the output contrast reaches 30.0 dB, indicating that the NOT can also be well fitted in this

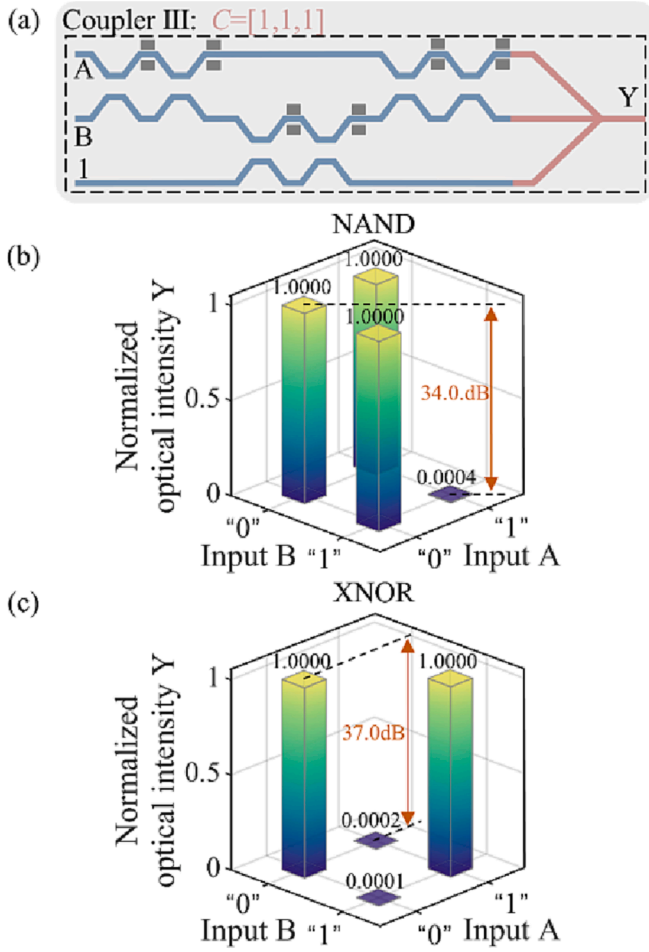


Fig. 5. Simulation results of the NAND and XNOR operations with a three-port OIU. (a) The configuration of OIU and coupler. (b) The output normalized optical intensity of (b) NAND and (c) XNOR operations.

configuration. Results shown in Fig. 4(d) indicate that the NOR logical operation has a similar intensity distribution to AND operation, for the input logic states “00,” “01,” “10,” and “11,” the output intensities are 1.0000, 0.0994, 0.0995, 0.1459, respectively, and the minimum contrast is 8.4 dB, which is slightly less than 9.3 dB of AND.

Furthermore, we consider the implementation of NAND and XNOR. Fig. 5 demonstrate the simulation results of the NAND and XNOR logic operations implemented by OIU with the coupler layer setting of $C = [1, 1, 1]$. In this case, all the outputs of the OIU are interfered with by the coupler to realize the logic operation. As aforementioned, this framework is determined by the desired logical operations (the square modulus of the vector A). For NAND and XNOR operations, the modulo square of vector A in functions $Y = AX$ is three, indicating that three unit vectors must be superimposed linearly. Fig. 5(b) and (c) illustrate the normalized output intensities of NAND and XNOR, respectively. It can be observed that all logic states “1” have the same intensity of 1, the logic state “0” is less than 0.0005 different from the ideal value, and the contrast ratios reach 34.0 dB and 37.0 dB.

3.2. Implementation of combinatorial logic operation

For AND and NOR operations, the output logic state “1” fits well, but the result for output logic state “0” differs from the ideal result. Therefore, we perform the modified AND operation by cascading NAND with NOT and perform the modified NOR operation by cascading OR with NOT, which is expected to improve the contrast ratios of logic states. As shown in Fig. 6(a) and (b), we realize this by connecting the output ports

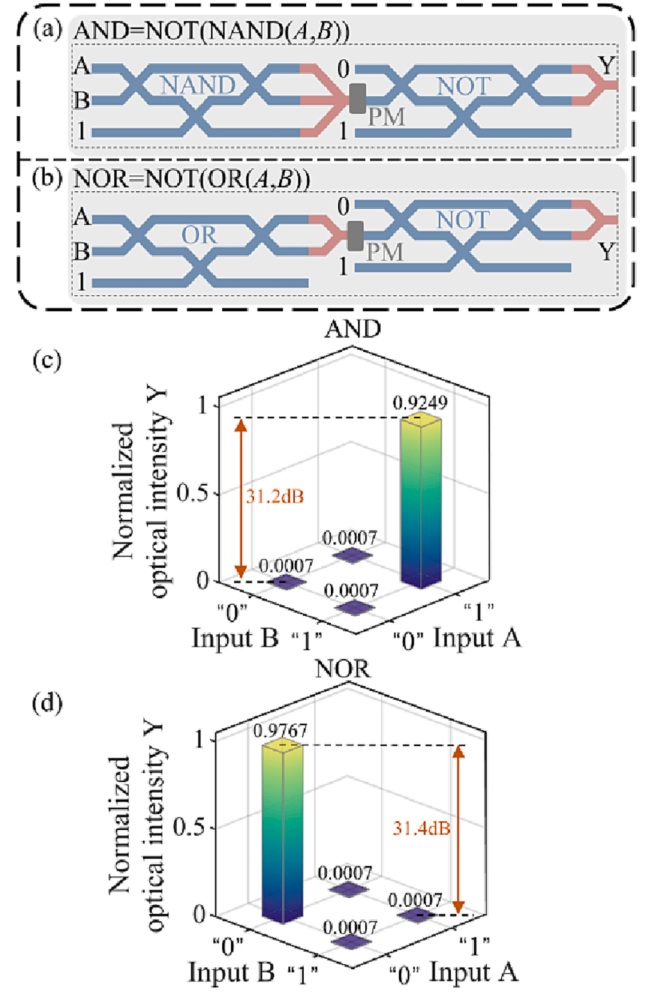


Fig. 6. The results of AND and OR operations by cascading NAND and NOT and cascading OR and NOT. (a) and (c) are the structure and output normalized intensity of AND, (b) and (d) are the structure and output normalized intensity of the structure of NOR. PM represents phase modulator.

of NAND and OR with the input port of NOT through a phase modulator (PM) that can maintain the phase matching of the input light to the next logic operation. Because the output intensity of the logic operation was consistent with the input intensity, no additional optical amplitude modulation was required. Fig. 6(c) and (d) show the corresponding logic operation results. Compared with Fig. 3(b) and 4(c), the output intensity of input states “00,” “01,” and “10” are significantly reduced, and the error with 0 is 0.0007. Although the intensities of output state “1” of modified AND and NOR operations decrease to 0.9249 and 0.9767, respectively, the output contrast exceeds 30 dB. Here, the third constant input “1” of NOT is used as the reference for phase modulation. In more complicated cases, a specific port needs to be used as a reference, and the phases of the other ports need to be adjusted based on the reference to achieve the phase matching. Note that the phase modulator here is also based on the lossless assumption. In practice, we can introduce the loss of the phase modulator into the model and re-optimize the global phase shift parameters through method such as in-situ optimization to achieve the maximum extinction ratio. These results demonstrate that logic operations based on OIUs can be cascaded without additional optical gain or attenuation but only with a phase modulator.

In addition, complex logic operations can be implemented by combining multiple-basis logic operations. Because the half adder does not require to consider carry from the low order, we only calculate the sum of the two binary inputs A and B and return two outputs, including

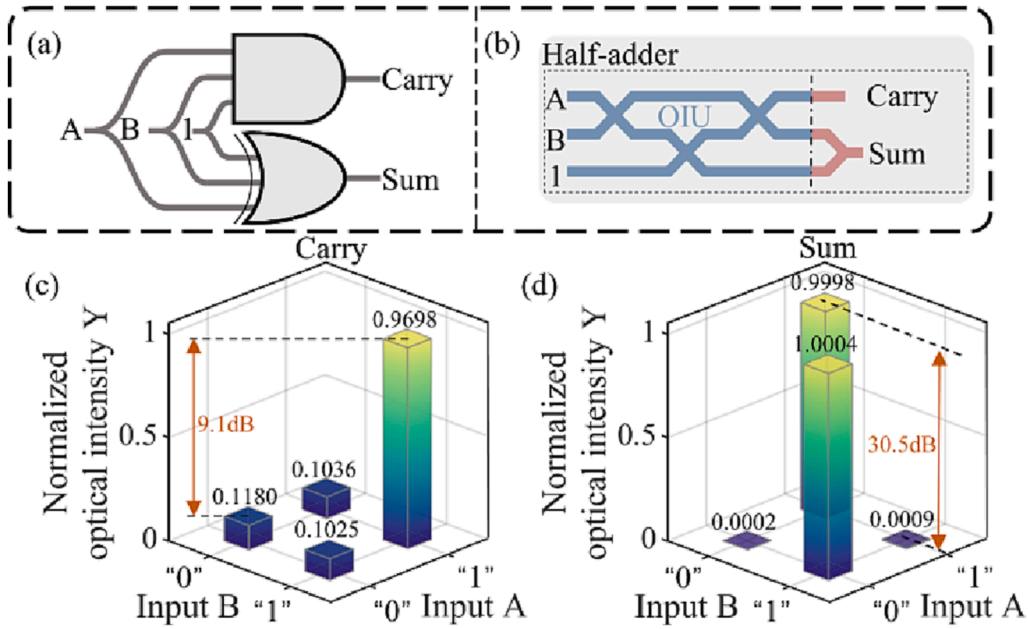


Fig. 7. The results of a half adder. (a) the half adder implemented by combining AND and XOR operations in parallel. (b) Multi-logic OIU scheme. (c) Output normalized intensity of Sum. (d) Output normalized intensity of Carry.

carry and sum. Mathematically, its response mechanism is the same as that of AND and XOR. Based on this principle, we constructed a half adder by combining the AND and XOR operations in parallel, as shown in Fig. 7(a). In this structure, inputs A and B are first split into two sub-beams in a 50:50 ratio, one of which is fed to the XOR for the standard sum operation, and the other is received by the AND for computing the carry. Because the entire logic operation is based on linear optics, the beam-splitting operation in the cascade results in a halving of the input amplitude, that is, each logic operation has an input amplitude of 0.5 or 0. Therefore, based on the previous analysis, for input states "00," "01," "10," and "11," the current output intensity of Carry is 0.0287, 0.0267, 0.0267, and 0.2461, and Sum is 0.0002, 0.2500, 0.2499, and 0.0002, respectively. Although the logic contrast remains unchanged after cascading, the output intensity is reduced by a factor of four (the output amplitude is reduced by a factor of two). Therefore, to maintain the overall consistency for mutually operating the outputs of each logic operation, the amplitude of the input must be increased to twice the normalized amplitude, or the output result must be normalized again.

To solve the problem of halving the output intensity caused by the beam-splitting operation, we describe another solution that uses a single OIU with multiple couplers to simultaneously accomplish the target operation. As shown in Fig. 7(b), the definition of the input port of the OIU is the same as before; the first output port is defined as Carry, and the second and third ports interfere through the coupler as Sum. Then, the equation can be written as follows:

$$\begin{aligned} \text{Carry} &= C_1 UX, \\ \text{Sum} &= C_2 UX \end{aligned} \quad (7)$$

where $C_1 = [1, 0, 0]$ and $C_2 = [0, 1, 1]$. Carry and Sum share the same U . The objective function simultaneously measures the loss value of Sum and Carry and is used to guide the optimization of the phase shift. Owing to the introduction of an optimization algorithm to solve the phase-shift parameters in the OIU, we can trade off the loss values between multiple different outputs to determine the optimal solution. The loss curve is shown in Fig. S3. Fig. 7(c) and (d) show the normalized output intensities of the optimized Carry and Sum, respectively. It can be observed that the output results are in agreement with Fig. 3(b) and 4(e). The contrast ratios of Carry and Sum were 9.1 dB and 30.5 dB, respectively. In addition, the intensities of the logic "1" of the two outputs were

0.9698 and (0.9998, 1.0004), respectively, which does not have the problem of four times the light intensity attenuation, as aforementioned.

As shown in the results here, besides phase modulation, the scalability of the proposed basic logic operations to implement combinatorial logic operations is mainly limited by two factors, one is the accumulation of operational errors and the other is the loss of optical power. On the one hand, because we used the gradient descent algorithm to approximately solve the Eq. 4, limited by the calculation accuracy, we can only obtain numerical approximate solutions rather than analytical solutions. (Even if we could solve the phase shifters analytically in some situations, it would become inefficient and difficult as the dimensions increases.) The slight differences in output intensities will propagate and gradually increase after passing through several logic blocks, eventually resulting in the indistinguishability of the output "0" and "1" logic states. On the other hand, in practice, non-ideal situations such as processing errors will cause device defects and lead to loss of optical power, which further limits the number of scalable blocks. For the two factors mentioned earlier, we can mitigate the errors arising from the cascade by making corrections to the MZI matrix to consider factors like insertion loss) and solving the analytical solution of the matrix equation. Alternatively, the scalability of the MZI array may provide us with another solution, that is, we can parallelize the inputs of each stage and use them as the input of a new logic block, thus realizing equivalent logic operation.

4. Discussion

In this study, we introduce a meshed MZI array to provide a linear complex amplitude modulation solution for logic operations. The seven basic logic operations were well implemented by an OIU with three MZIs in the simulation. Compared with previous linear modulation schemes [29,34,35,40], we transformed the amplitude attenuation (gain), phase modulation, and polarization modulation of external devices into phase-only modulation inside the MZIs. Furthermore, the normalized intensity characteristic of logic states "0" and "1" were guaranteed, so that the operation could be directly interconnected with only a phase modulation operation. More importantly, the previous plasmonic and MZI-based switching devices, such as optical router, wavelength demultiplexer, and logic gates, are usually rely on design methods based on

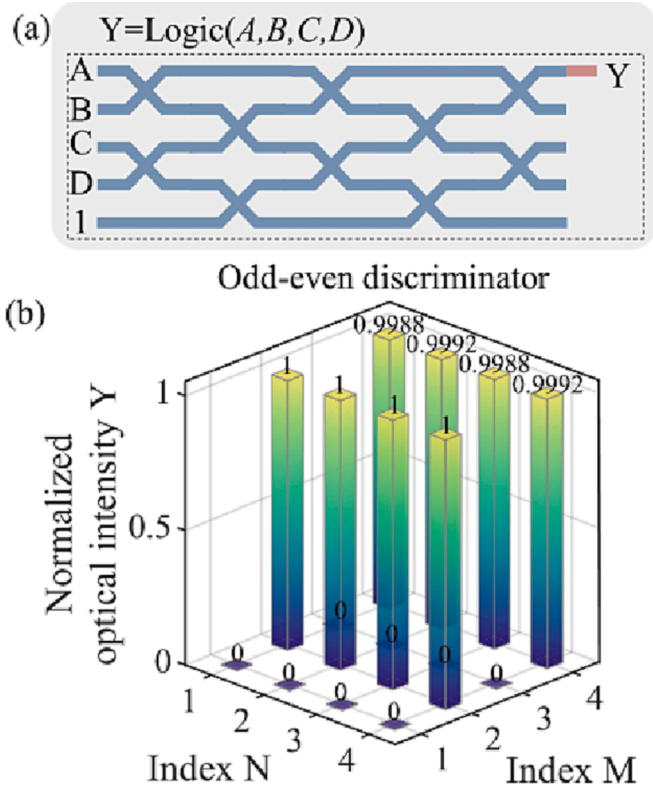


Fig. 8. The results of a 4-bit odd number discriminator. (a) The structure of a 5-port OIU that implements 4-input logic operations. (b) The output intensity distribution of the optimized discriminator.

fixed empirical paradigms, and thus not flexible enough to implement complex logic operations [31–35,41–45]. The proposed MZI array can solve the required phase shifter parameters for different logic operations in a target-oriented manner through gradient descent algorithms. Therefore, it allows us to implement different optical logic operations using the same architecture, freeing us from relying on empirical paradigms. In addition, we can implement logic operations with more than two inputs by extending the size of the OIU; for example, a 4-bit odd–even number discriminator that determines whether an input 4-bit binary number is odd, if it is odd, returns $Y = “1”$, otherwise returns $Y = “0”$. Fig. 8(a) shows the structure of a 5-port OIU that implements 4-input logic operations. Fig. 8(b) shows the normalized intensity distribution of the output of the optimized discriminator. For the sake of intuition, the x- and y-axes are two indices, M and N , respectively, and their coordinates (M, N) indicate that the input value here is $X = M + 4(N - 1) - 1$. For example, coordinate $(2, 3)$ indicates that the input number is $X = 9$, which corresponds to $(A, B, C, D) = (0, 1, 0, 1)$, etc. The z-axis represents the discriminant result, that is, the logic output $Y = \text{Logic}(X)$, which is the normalized intensity. It can be observed that the 4-bit binary number composed of (A, B, C, D) can be discriminated well. For example, when the input was $(A, B, C, D) = (1, 0, 1, 0)$, the normalized intensity of the output was zero, indicating that it was even. When the input was $(A, B, C, D) = (1, 0, 1, 1)$, the normalized intensity of the output was 0.9992, indicating that it was odd. Furthermore, there is a perfect ratio between logic “1” and “0”, that is, the contrast is infinite.

Owing to their linear transformation properties, logic operations implemented using OIUs can tolerate fluctuations in the input optical intensity (amplitude) within a certain range. Fig. 9 shows the relationship between the output and input of the seven optical logic operations. Note that the x-, y-, and z-axes are all expressed in terms of normalized amplitudes, and they correspond to inputs A, B, and output Y, respectively. It can be observed that the change between them satisfies a linear relationship, where the amplitude of the output Y changes synchronously with that of the inputs A and B. Because we are essentially

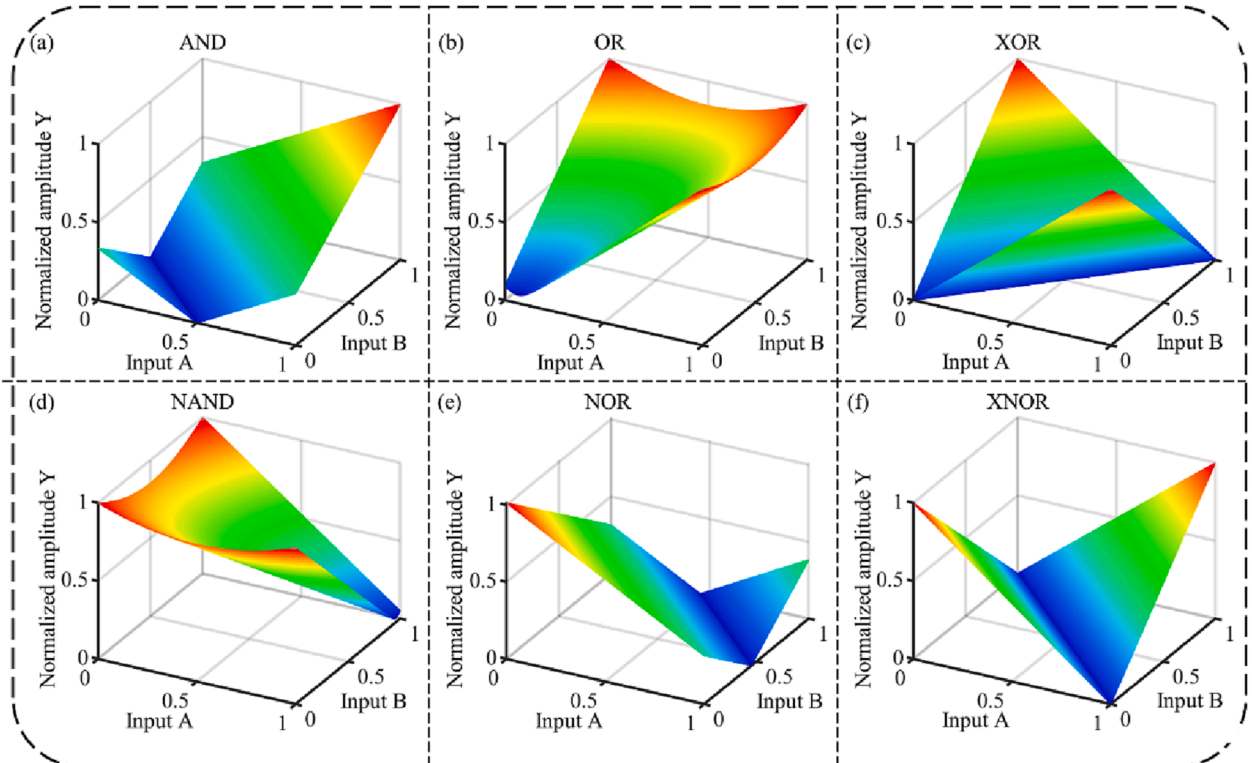


Fig. 9. The relationship between output and input of the seven basic optical logic operations. The x- and y-axes represent the normalized amplitude of the inputs A and B, respectively, and the z-axis represents the normalized amplitude of the output Y. (a)–(f) correspond to AND, OR, XOR, NAND, NOR, and XNOR, respectively.

operating on normalized amplitudes here (and not optical intensity), for any logic gate we can define the maximum amplitude of the input to be the logic state “1”, and the smallest to be the logic state “0”. Therefore, by defining an intermediate amplitude M , we can transform the problem of logic operations into a classification problem and judge the output amplitude greater than M as a logic state “1”, and less than as a “0”. Therefore, as shown in Fig. 9, we can define the output below 0.5 as the logic state “0” and the output greater than 0.5 as the logic state “1”. This does not change the final logic operation result, and provides a more relaxed decision margin. Thus, based on the advantages of a linear transformation, we endow the logic operation with tolerance to the input and output amplitudes, thereby alleviating the requirement for precise light control.

In this work, we designed and investigated different logic operations using unitary matrix–vector multiplication of MZI array based on the lossless assumption of MZIs. This assumption is advantageous and valid. It allows us to focus on and explore the fundamentals of linear-optical logic operations without getting too entangled in the specific implementation devices. However, it also makes us ignore metrics such as insertion loss, which is caused by component imperfections and are related to specific waveguide parameters. It affects the extinction ratio. Fortunately, once the proposed MZI array is machined with proper waveguide parameters, the total insertion loss can be practically measured by comparing the light intensity of all input and output ports of MZI array. And it can be used to modify the lossless MZI matrix and combined with techniques such as in-situ optimization to achieve close to ideal contrast ratio in practice [37,38].

5. Conclusion

In this study, we introduce a meshed MZI array to provide a linear complex amplitude modulation solution for optical logic operations. The optical interference unit OIU is formed by a meshed MZI array, and the unitary matrix–vector product operations are imitated. Owing to the introduction of linear vector composition, the phase-shift parameters of the OIU can be optimized using a gradient descent algorithm to perform non-unitary logic operations. We show that the seven basic logic operations are well implemented by an OIU with three MZIs in the simulation, and take AND, NOR, and half-adder as examples to illustrate the two connection modes of logic operations, namely sequential and parallel connections. In addition, the potential of the OIU to construct multi-logic devices was demonstrated. Because these logic operations are implemented on the same platform, this strategy provides a promising method for device fabrication and system integration, which is of great significance in optical digital computing and signal processing.

CRedit authorship contribution statement

Jiafu Chen: Software, Investigation, Data curation, Writing – original draft, Methodology. **Linwei Tang:** Software, Investigation, Data curation. **Zebin Huang:** Investigation, Data curation, Writing – review & editing. **Wenjie Xiong:** Investigation, Data curation. **Junmin Liu:** Data curation, Formal analysis. **Peipei Wang:** Investigation, Data curation, Writing – review & editing. **Ying Li:** Resources, Funding acquisition. **Dianyuan Fan:** Supervision, Funding acquisition. **Shuqing Chen:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.optlastec.2023.109703>.

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